

Advanced (Habanero)

Q1. Factor the expression: $x^2 + 5x + 4x + 20$

- (A) $(x + 4)(x + 5)$
- (B) $(x + 2)(x + 10)$
- (C) $(x + 1)(x + 20)$
- (D) $(x + 3)(x + 15)$
- (E) $(x + 5)(x + 4)$

Q2. Factor the expression: $3x^2 + 12x + 4x + 16$

- (A) $3(x + 4)(x + 2)$
- (B) $(3x + 4)(x + 4)$
- (C) $(3x + 2)(x + 4)$
- (D) $3(x + 2)(x + 4)$
- (E) $(3x + 4)(x + 2)$

Q3. Factor the expression: $x^2 - 7x - 2x + 14$

- (A) $(x - 2)(x - 7)$
- (B) $(x - 7)(x - 2)$
- (C) $(x + 2)(x - 7)$
- (D) $(x - 7)(x + 2)$
- (E) $(x - 2)(x + 7)$

Q4. Factor the expression: $x^2 + 10x - 3x - 30$

- (A) $(x + 10)(x - 3)$
- (B) $(x - 3)(x + 10)$
- (C) $(x + 5)(x - 6)$
- (D) $(x - 6)(x + 5)$
- (E) $(x + 6)(x - 5)$

Q5. Factor the expression: $2x^2 + 5x + x + 10$

- (A) $(2x + 5)(x + 2)$
- (B) $(2x + 2)(x + 5)$
- (C) $(x + 5)(2x + 2)$
- (D) $(x + 2)(2x + 5)$
- (E) $(2x + 10)(x + 1)$

Q6. Factor the expression: $3x^2 - x - 12x + 4$

- (A) $(3x - 1)(x - 4)$
- (B) $(3x + 1)(x - 4)$
- (C) $(3x + 4)(x - 1)$
- (D) $(3x - 4)(x + 1)$
- (E) $(3x + 4)(x - 1)$

Q7. Factor the expression: $x^2 - 4x - 5x + 20$

- (A) $(x - 4)(x - 5)$
- (B) $(x + 4)(x + 5)$
- (C) $(x + 5)(x - 4)$
- (D) $(x + 4)(x - 5)$
- (E) $(x - 4)(x + 5)$

Q8. Factor the expression: $2x^2 + 8x + 5x + 20$

- (A) $2(x + 4)(x + 5)$
- (B) $(2x + 5)(x + 4)$
- (C) $2(x + 5)(x + 4)$
- (D) $(x + 5)(2x + 4)$
- (E) $2(x + 4)(x + 5)$

Open Ended Questions (FRQ)

Q9. Factor the expression: $x^2 + 2x - 6x - 12$

Q10. Factor the expression: $3x^2 - 9x + 2x - 6$

Chef's Tips

- Tip 1: Check if the first and last terms have any common factors
- Tip 2: Look for common binomial factors in the middle terms
- Tip 3: Verify factoring by multiplication